

Design and control of a single-stage dual-actuator system for high-precision manufacturing

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Abstract This paper proposes a design for the linear state feedback control of the dual-actuator system, which is a dual-input single-output system for the high-precision manufacturing stage. The proposed control prevents saturation or reduces the unnecessary movement of the piezoelectric (PZT) actuator at the transient response by tracking the error between the estimated and actual positions of the coarse-actuator system at each control sample. Also, a new mechanism of the single-stage dual actuator is introduced. The axes of the stepper motor and the PZT actuator are co-axial. The coupling effects between the stepper motor and the PZT actuator are considered. Both the simulation and experiment results show that the proposed algorithm successfully prevents unwanted motions of the PZT actuator at the transient response. The experiment results show that the settling time and overshoot were enhanced by 45.7 and 95.9 %, respectively, for the proposed algorithm when the reference distance was 10 μm , which exceeds the stroke of the PZT actuator.

1 Introduction

The dual-stage actuator system has been introduced to control a system with high precision while achieving a long servo distance. A coarse actuator can travel over a long distance, but it has poor accuracy. A fine actuator is suitable for high-precision positioning, and its response is fast, but its stroke is limited. In general, the voice coil motor is used as a coarse actuator, and the piezoelectric (PZT)

actuator is used as a fine actuator, for the hard disk drive (HDD) system. For the linear stage system, in general, a linear-motor- or ball-screw-driven system is used as a coarse-actuator system, and the PZT actuator is used as a fine actuator. Therefore, by combining these two actuators within the same axis, the system will be able to travel over a long servo distance in nanoscale resolution. The dual-stage actuator system has been widely used in a variety of applications as electronic parts require micro- or nanoscale processing or manufacturing. Its mechanism has been implemented in HDDs for high tracking density (Hu et al. 1999; Kobayashi and Horowitz 2001; Suh et al. 2002; Li et al. 2011). Also, it has been used in micro-manipulators (Kwon et al. 2000), positioning tables (Yao and Tomizuka 1993), machining systems (Kim et al. 2004), and high-resolution manufacturing stage systems (Zheng et al. 2010; Salton et al. 2011; Kim et al. 2011).

The dual-stage actuator system has two actuators, with one feedback encoder in a single axis. In other words, the system exhibits the dual-input single-output (DISO) system. In general, the coarse actuator has a low-frequency response while the fine actuator has a high-frequency response. When the system tracks a desired distance, the coarse and fine actuators are mutually operated, and each actuator affects the other's motion. If a controller is designed to reject disturbances, a coarse actuator rejects low-frequency disturbances and a fine actuator rejects high-frequency disturbances. Therefore, it is important to design a controller that can achieve good performance in both time and frequency response, to prevent the negative mutual operation of the two actuators. A number of control algorithms have been studied for the dual-stage actuator system. In many industrial applications, the controller is designed in such a way that a coarse actuator tracks the desired distance while a fine actuator is used to compensate

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for the position error (Koganezawa et al. 1999). In this approach, however, the mutual effects of the two actuators are not taken into account, and it is hard to expect a good frequency response. The PQ method and the sode plot are studied to define the DISO and single-input single-output (SISO) systems for reducing the control design problem (Schroek et al. 2001). This approach stresses the effect of the open-loop frequency function on the closed-loop sensitivity function, and stresses output allocation with regard to the frequency function (Rapley and Messner 2001).

The track-following design has been studied for the dual-stage actuator system (Semba et al. 1999; Hernandez et al. 1999). High controller bandwidth has been achieved and developed based on the frequency characteristics of the fine- and coarse-actuator paths. Only the step response of the closed-loop track-seeking performance of dual-stage actuators, however, has been addressed. To address this issue, a decoupled track-following controller was developed (Kobayashi and Horowitz 2001). The decoupled track-following controller shows improved overshoot and improved following performance. The coarse and fine actuators are mutually operated, and each actuator affects the other's motion in the dual-stage actuator system, as mentioned earlier. Therefore, it is important to define the coupled dynamics between the two actuators for enhanced transient response. In many studies, however, the coupled dynamics are neglected (Yao and Tomizuka 1993; Koganezawa et al. 1999; Schroek et al. 2001; Rapley and Messner 2001; Zheng et al. 2010). In some studies, the coupled dynamics are defined so that the controller would show an enhanced transient response in the time domain (Suh et al. 2002). A number of robust controllers have also been studied for stability and disturbance. A robust linear controller using the μ -analysis tool (Hernandez et al. 1999), H2 control (Callafon et al. 2006), and H-infinity control (Ding et al. 2000; Li and Horowitz 2001) has been used. These approaches successfully showed that it has reliable robust stability against the uncertainty and disturbance of the plant. Controllers using an estimator or observer have been studied to achieve a reliable response in both the time and frequency domains (Lee and Suh 2011). Feed-forward controllers have also been studied (Devasia 2007; Syahputra et al. 2012).

In previous studies, the PZT actuator showed unnecessary movement at the transient response, or it was saturated if the reference signal was beyond the PZT actuator stroke because the PZT actuator has fast response, and it tracks the error signal between the reference and the position of the system or between the reference and position of the coarse-actuator system (Hu et al. 1999; Kobayashi and Horowitz 2001; Zheng et al. 2010; Lee and Suh 2011; Li et al. 2011; Salton et al. 2011). This paper proposes a robust feedback control design for the DISO high-precision

manufacturing stage system that prevents saturation or reduces the unnecessary movement of the PZT actuator during the transient response. The PZT actuator tracks the error of the expected position of the stepper motor at each sampling time. In this method, the input signal to the PZT actuator can be minimized during the transient response. A robust controller was designed based on the LQG/LTR approach and an estimator system.

The mechanical structure in many studies has shown that two actuators are driven in the same axis, but both axes are not co-axial. The HDD system has a two-joint and two-link mechanism (Kobayashi and Horowitz 2001; Suh et al. 2002; Li et al. 2011). In the linear manufacturing system, there are two separate stages, and both are assembled vertically (Kwon et al. 2000). This mechanism structure can be an issue when many dual-stage actuator systems are combined for a multi-axis manufacturing system with nanoscale resolution because the deflection of the structure will affect the resolution. This paper proposes a mechanism with two actuators that are co-axial in a single stage. Therefore, it can be considered a single-stage dual-actuator system.

This paper is organized as follows. Section 2 introduces a new mechanism of the dual-actuator single-stage system, and discusses the modeling of the system. The proposed control algorithm is explained in Sect. 3. The simulation results are shown in Sect. 4, and the experiment results are presented in Sect. 5. The paper is summarized in Sect. 6.

2 System modeling

2.1 Mechanism of the single-stage dual-actuator system

The system studied in this paper has a new mechanism assembly, as aforementioned. The system has a ball screw mechanism driven by a stepper motor for long-distance travel, and has a PZT actuator for fine positioning. The travel range of the ball screw mechanism driven by a stepper motor is 30 cm, with 480 nm resolution, and the PZT actuator stroke is $\pm 16 \mu\text{m}$, with 0.1 nm open-loop sensitivity and 11 kN maximum load. Both actuator systems are operated in the same axis, and both axes are co-axial, making it different from the conventional system. The structure of the PZT actuator is that of a hollow cylinder, and it is installed between the support structure and the ball screw mechanism, as shown in Fig. 1. The PZT actuator positions the ball screw mechanism so that the stage is positioned in nano scale. Therefore, the system can be considered a single-stage dual-actuator system, as aforementioned. In this design approach, the volume of the system can be decreased with greater mechanical stability.

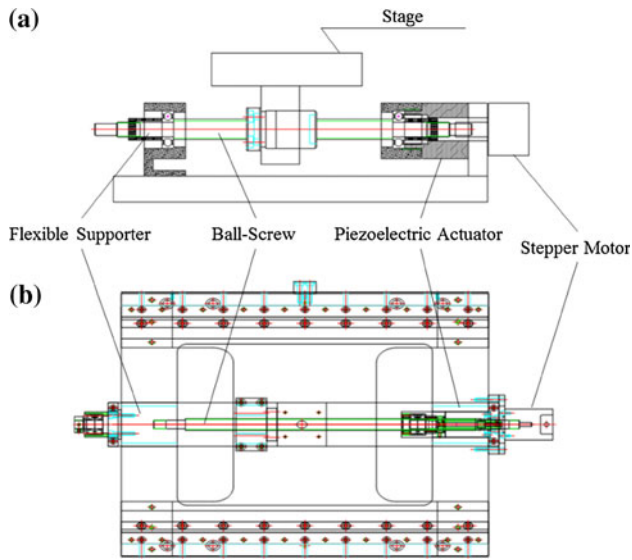


Fig. 1 Mechanical structure of the single-stage dual-actuator system: **a** side view; and **b** top view

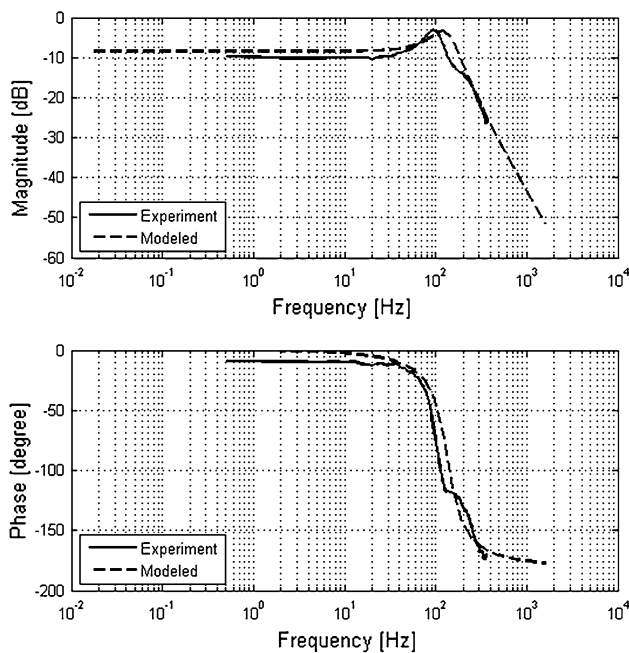


Fig. 2 Frequency response of the piezoelectric actuator

The system has one feedback system, which is an exposed linear encoder; thus, the system exhibits the DISO system.

2.2 Plant model

For high-precision control in nanometer scale, it is important to define the plant model accurately. In this study, actuator models for both the ball screw mechanism driven by the stepper motor and the PZT actuator were

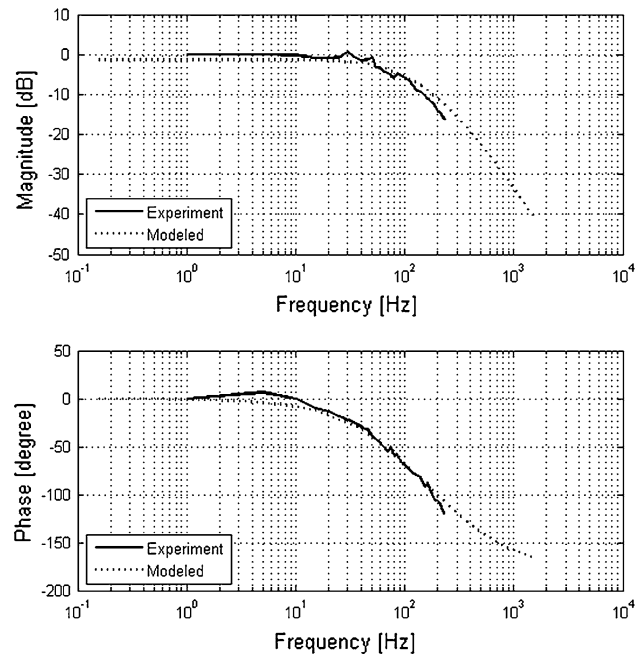


Fig. 3 Frequency response of the ball screw mechanism driven by the stepper motor

derived experimentally. The frequency responses of both plants were obtained via the sine sweep test. Figures 2 and 3 show the frequency responses of the PZT actuator and of the ball screw mechanism driven by the stepper motor, respectively. The fine-actuator system had a -3.188 dB frequency peak at 120.1 Hz and was modeled as a linear second-order system. Hence, the transfer function of the fine-actuator system can be represented as

$$G_{fine}(s) = \frac{K_{pf} \cdot \omega_{nf}}{s^2 + 2\zeta_f \omega_{nf} s + \omega_{nf}^2}, \quad (1)$$

where K_{pf} is the gain of the fine-actuator system, ω_{nf} is the system's natural frequency, and ζ_f is the system's damping coefficient.

The coarse-actuator system had a -2.104 dB frequency peak at 37.08 Hz and was modeled as a linear second-order system. The transfer function of the coarse-actuator system can be represented as

$$G_{stepper}(s) = \frac{K_{pc} \cdot \omega_{nc}}{s^2 + 2\zeta_c \omega_{nc} s + \omega_{nc}^2}, \quad (2)$$

where K_{pc} is the gain of the coarse-actuator system, ω_{nc} is the system's natural frequency, and ζ_c is the system's damping coefficient.

2.3 Dynamic model

As the plant models of both the PZT actuator and the ball screw mechanism were derived as linear second-order

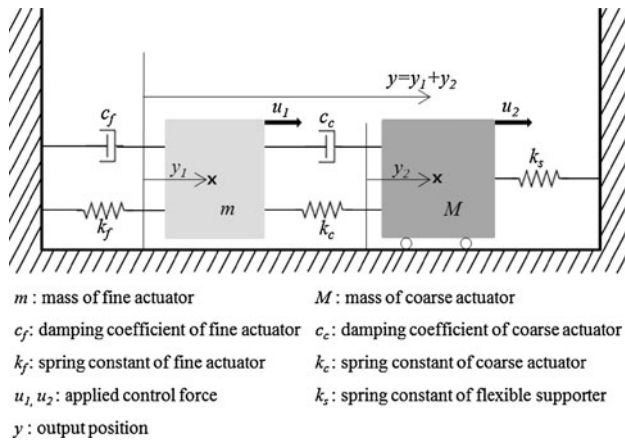


Fig. 4 Dynamic model of the system

systems, the single-stage dual-actuator system can be described as a second-order mass-spring-damper system, as shown in Fig. 4.

When the system is operated to track the desired distance, both the PZT actuator and the stepper motor are mutually operated. Each actuator system, however, can affect the other's motion. For high-precision control with good transient response, it is important to consider the coupled dynamics in the control algorithm. Applying Newton's second law to the two-degree-of-freedom control system, the state space representation of the system is as follows:

$$\begin{bmatrix} \dot{x}_f \\ \ddot{x}_f \\ \dot{x}_c \\ \ddot{x}_c \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -\frac{k_f+k_c}{m} & -\frac{c_f+c_c}{m} & -\frac{k_c}{m} & -\frac{c_c}{m} \\ 0 & 0 & 0 & 1 \\ \frac{k_c}{M} & \frac{c_c}{M} & -\frac{k_c+k_s}{M} & -\frac{c_c}{M} \end{bmatrix} \begin{bmatrix} x_f \\ \dot{x}_f \\ x_c \\ \dot{x}_c \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 1/m & 0 \\ 0 & 0 \\ 0 & 1/M \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}. \quad (3)$$

Transforming Eq. (3) into transfer matrix $G(s)$, the transfer function of the system is described as

$$G(s) = \begin{bmatrix} \frac{Y_1(s)}{U_1(s)} & \frac{Y_1(s)}{U_2(s)} \\ \frac{Y_2(s)}{U_1(s)} & \frac{Y_2(s)}{U_2(s)} \end{bmatrix}. \quad (4)$$

From the characteristics of the assembly structure, however, it is assumed that the ball screw mechanism's motion affects the PZT actuator motion but that the PZT actuator's motion does not affect the ball screw mechanism's motion. Therefore, $Y_1(s)/U_2(s) = 0$, and $Y_2(s)/U_1(s)$ is defined as coupled dynamics. The transfer function of the coupled dynamics is

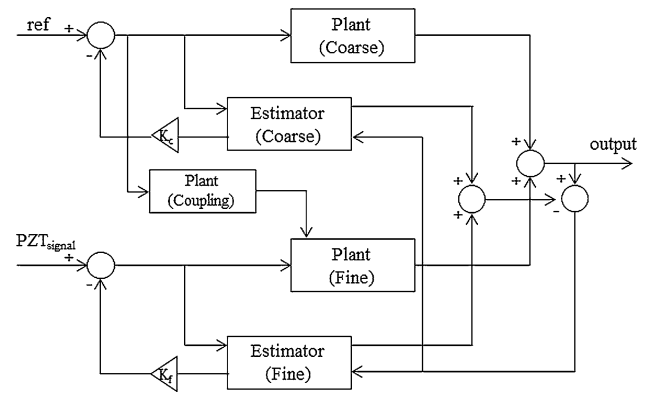


Fig. 5 Control structure of the system

$$G_{coup}(s) = \frac{(c_c s + k_c)/Mm}{s^4 + a_1 s^3 + a_2 s^2 + a_3 s + a_4}, \quad (5)$$

where

$$\begin{aligned} a_1 &= ((c_f + c_c)/m + c_c/M); \\ a_2 &= ((k_f + k_c)/m + (k_c + k_s)/M + c_c(c_f + c_c)/Mm); \\ a_3 &= (c_c(k_f + k_c) + (c_f + c_c) \times (k_c + k_s))/Mm; \text{ and} \\ a_4 &= (k_f + k_c)(k_c + k_s)/Mm. \end{aligned}$$

3 Controller design

3.1 Dual-actuator control algorithm

The control algorithm for the DISO single-stage dual-actuator system is proposed in this paper. The PZT actuator tracks the coarse-actuator position error signal, which is an error between the estimated and actual positions of the coarse actuator at each control sample. In many dual-stage actuator studies, however, the PZT actuator follows PES, which is an error between the reference and the output position (Li and Horowitz 2001), or VPES, which is an error between the reference and the position of the coarse actuator (Suh et al. 2002). The PZT actuator showed unnecessary operation or was saturated at the early stage of the transient response for such control algorithms. In this paper, saturation or unnecessary movement of the PZT actuator can be prevented at the transient response by tracking

$$PZT_{signal}(k) = C_c \hat{x}_c(k) - (y(k) - C_f \hat{x}_f(k)), \quad (6)$$

where $y(k)$ is the output of the dual-actuator system, and $\hat{x}_{f,c}$ is the current estimate of the fine and coarse actuators, respectively.

The proposed control algorithm is shown in Fig. 5. The coupling effect was taken into account, and estimators were designed for both plants. In addition, the proposed

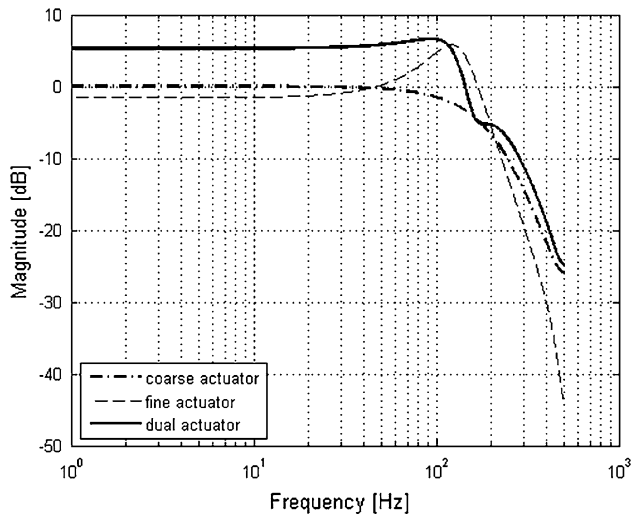


Fig. 6 Open-loop transfer functions

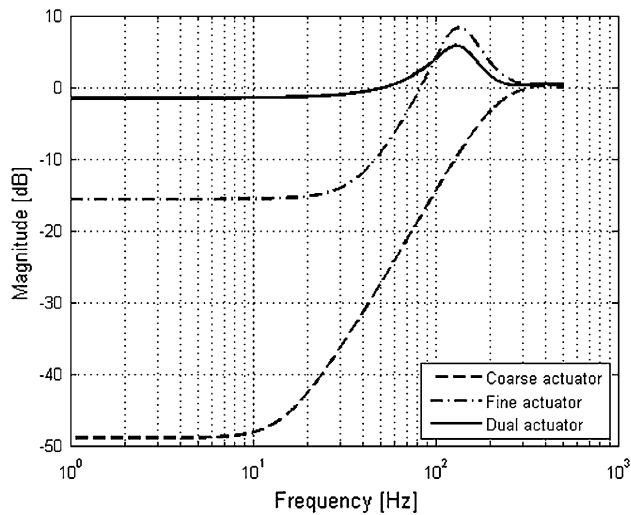


Fig. 7 Sensitivity functions

algorithm allows the PZT actuator to settle down to zero at the steady state so that the actuator could respond to the sequential step responses.

3.2 State feedback controller design

A current estimator was designed for both plants, to implement state feedback control and to derive the PZT_{signal} .

$$\hat{x}_{f,c}(k) = \bar{x}_{f,c}(k) + L_{f,c}[y(k) - C_{f,c}\bar{x}_{f,c}(k)], \text{ and} \quad (7)$$

$$\begin{aligned} \bar{x}_{f,c}(k+1) = & A_{f,c}\bar{x}_{f,c}(k) + B_{f,c}u_{f,c}(k) \\ & + A_{f,c}L_{f,c}[y(k) - C_{f,c}\bar{x}_{f,c}(k)], \end{aligned} \quad (8)$$

where $\bar{x}_{f,c}(k)$ is the predicted estimate based on a model prediction from the previous time estimate. The pole

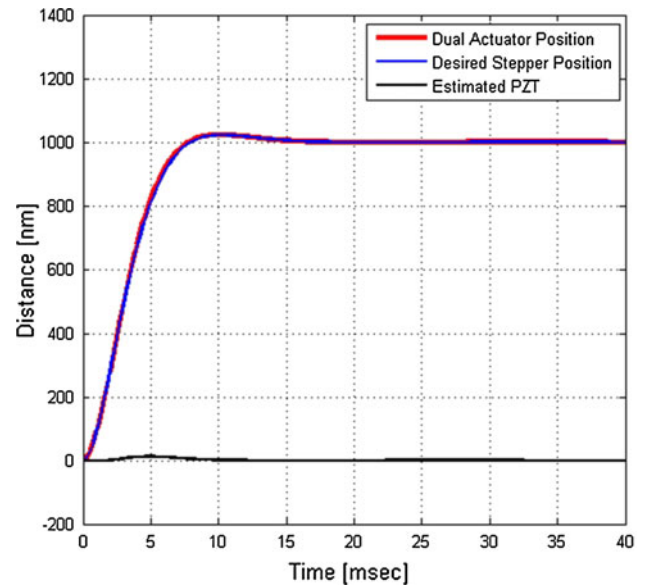


Fig. 8 Simulation result of the proposed algorithm

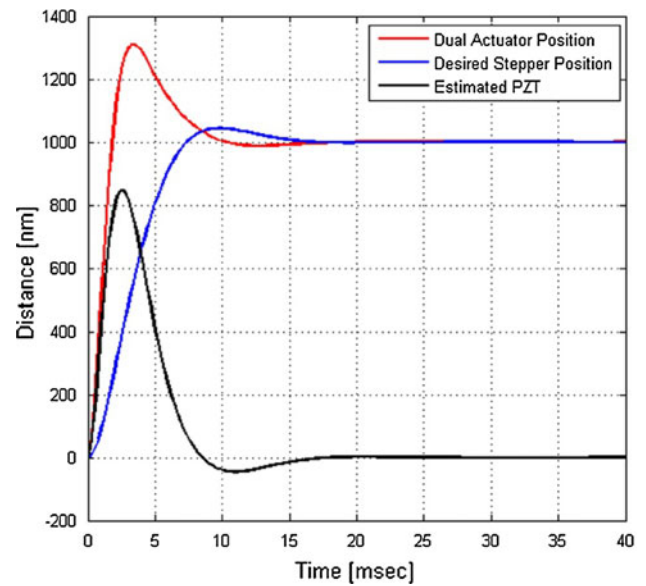


Fig. 9 Simulation result of the conventional algorithm (Suh et al. 2002)

placement method is used to define the current estimator gain $L_{f,c}$, to improve the transient response. The desired poles of the coarse-actuator system are $-66.623 \pm 66.64i$, and those of the fine-actuator system are $-177.66 \pm 177.72i$. Thus, the estimator gain is

$$L_c = \begin{bmatrix} -0.4434 \\ 267.7173 \end{bmatrix},$$

$$L_f = \begin{bmatrix} 0.6114 \\ -127.9435 \end{bmatrix}.$$

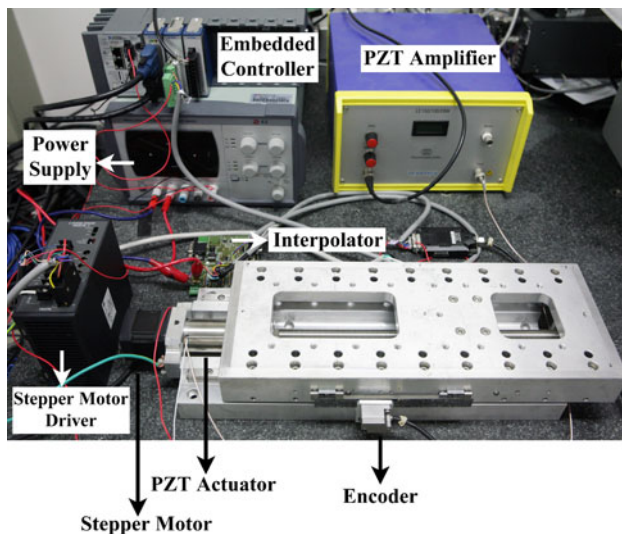


Fig. 10 Experiment set-up of the single-stage dual-actuator system

The state feedback control law is

$$u_{cf}(k) = -K_{cf} \cdot \hat{x}_{cf}(k). \quad (9)$$

Feedback control gain K_{cf} is defined as follows, to minimize the cost function by discrete-time LQR design:

$$J = \sum_{k=0}^N (x_k^T Q x_k + u_k^T R u_k). \quad (10)$$

The feedback control gain is

Fig. 11 System control architecture

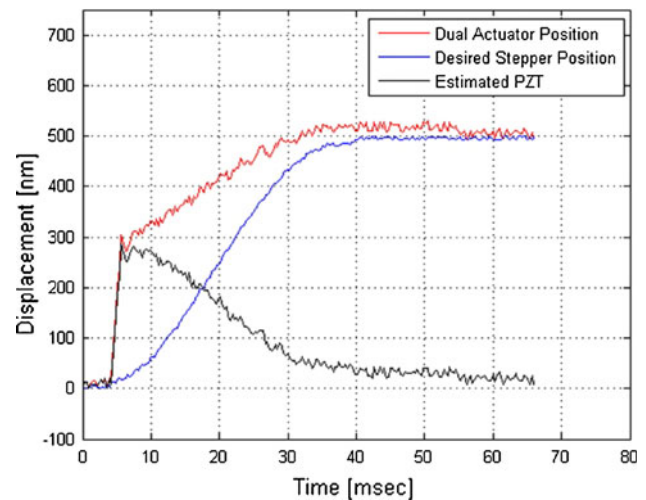
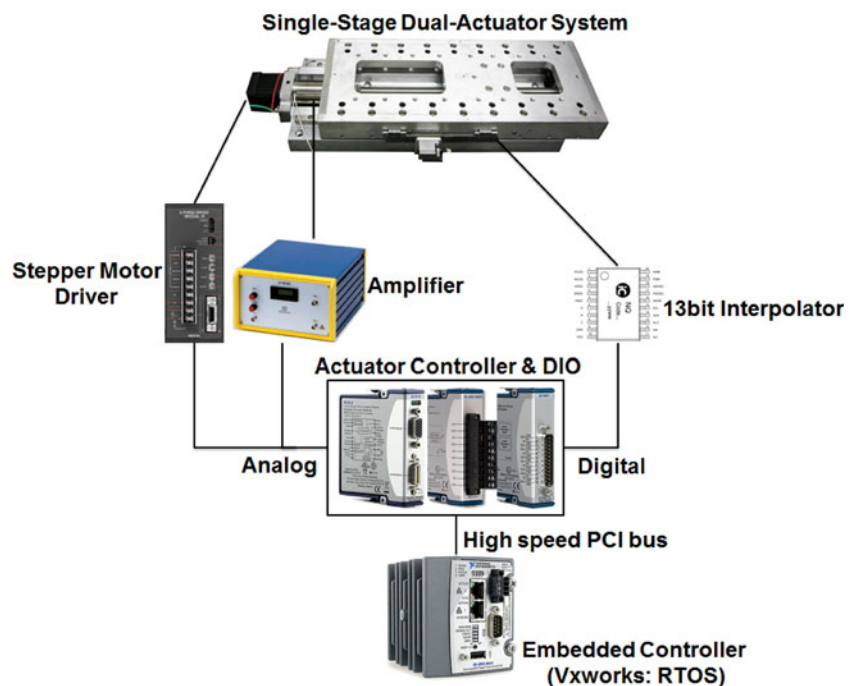


Fig. 12 Experiment results of the conventional algorithm when the reference distance was 500 nm

$$K = (R + B^T P B)^{-1} B^T P A, \quad (11)$$

where P is the unique positive solution to the discrete-time algebraic Riccati equation.

$$P = Q + A^T (P - P B (R + B^T P B)^{-1} B^T P) A \quad (12)$$

The calculated feedback gains are

$$K_c = [-0.9486 \quad -0.0034], \text{ and}$$

$$K_p = [-0.9004 \quad -0.0006].$$

The controller includes the control feedback. Using (7) yields

$$\hat{x}_{c,f}(k) = [A_{c,f} - B_{c,f}K_{c,f} - L_{c,f}C_{c,f}A_{c,f} + L_{c,f}C_{c,f}B_{c,f}K_{c,f}]\hat{x}_{c,f}(k-1) + L_{c,f}y(k) \quad (13)$$

For servo tracking, control law (9) becomes

$$u_{c,f}(k) = -K_{c,f}\hat{x}_{c,f}(k) + \bar{N}_{c,f} \cdot ref_{c,f}, \quad (14)$$

where $\bar{N}$ is the command matrix, ref_c is the desired distance of the system, and ref_f is the PZT_{signal} . To analyze the response of the system, estimator equations have to be combined with the system model for control (15). Combining (4) and (13), the system equations of the reference inputs with the current estimator are

$$\begin{aligned} \begin{bmatrix} x_{c,f}(k+1) \\ \hat{x}_{c,f}(k+1) \end{bmatrix} &= \begin{bmatrix} A_{c,f} & -B_{c,f}K_{c,f} \\ L_{c,f}C_{c,f}A_{c,f} & A_{c,f} - B_{c,f}K_{c,f} - L_{c,f}C_{c,f}A_{c,f} \end{bmatrix} \\ &\times \begin{bmatrix} x_{c,f}(k) \\ \hat{x}_{c,f}(k) \end{bmatrix} + \begin{bmatrix} B_{c,f}\bar{N}_{c,f} \\ B_{c,f}\bar{N}_{c,f} + L_{c,f}C_{c,f}B_{c,f}\bar{N}_{c,f} \end{bmatrix} r(k)_{c,f} \end{aligned} \quad (15)$$

where $r_c(k)$ is the desired position of the system, and $r_f(k)$ is the $PZT_{signal}(k)$. For (15), it is assumed that there is no disturbance to the system (Figs. 6, 7).

4 Simulations

Figures 8 and 9 show the simulation results of the proposed algorithm and of the algorithm studied by Suh et al. (2002), respectively. One micrometer, which is approximately 3 % of the PZT actuator stroke, was chosen for the reference signal, to check the response of the fine actuator at the transient response. The simulation results show that the movement of the PZT actuator decreased by 95 % while tracking the reference, with no steady-state error. Therefore, the unnecessary movement of the fine actuator at the early stage of control was greatly decreased, resulting in a faster settling time and less system overshoot.

5 Experiment and results

5.1 Experiment set-up

The experiment set-up of the single-stage dual actuator is shown in Fig. 10. National Instrument compactRIO, which is run by the Vxworks real-time operating system (RTOS) and field-programmable gate array (FPGA), was used for the main controller. For the feedback system, Heidenhain

linear exposed encoder LIP401 and an iC-NQ 13-bit signal-conditioning interpolator chip were used. The encoder generates sine and cosine waves with phase delay at 90° , whose signal periods are $2 \mu\text{m}$. The iC-NQ interpolator chip was set to 9 bits. Therefore, the resolution was 3.9063 nm . The HPSt 150/20-15/25 VS35 PZT actuator and the LE 150/100 PZT amplifier were used for nano-scale positioning, and the VEXTA RK566 stepper motor was used for long-distance positioning (Fig. 11).

5.2 Experiment results

Figures 12 and 13 show the experiment results of the conventional and proposed algorithms, respectively, when the reference distance was within the PZT actuator stroke.

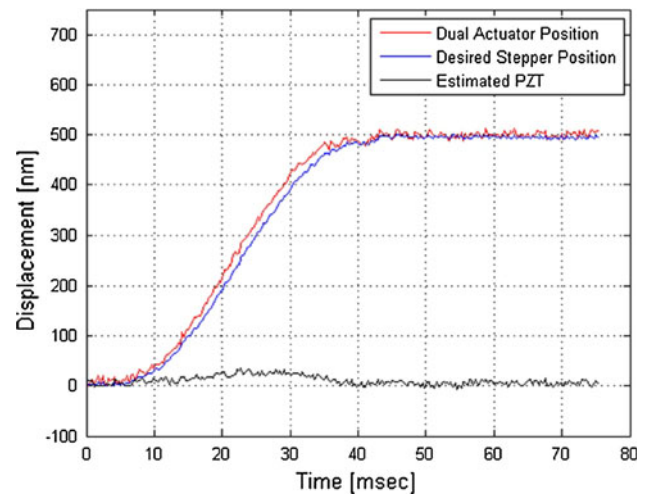


Fig. 13 Experiment results of the proposed algorithm when the reference distance was 500 nm

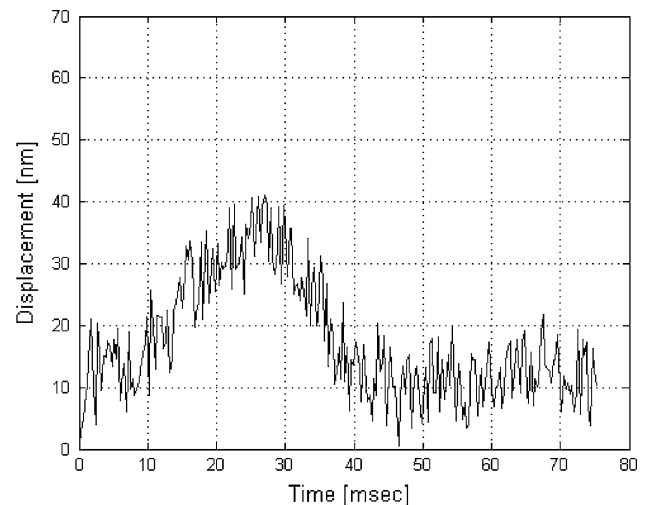


Fig. 14 Estimated position of the PZT actuator when the reference distance was 500 nm (estimated PZT from Fig. 13)

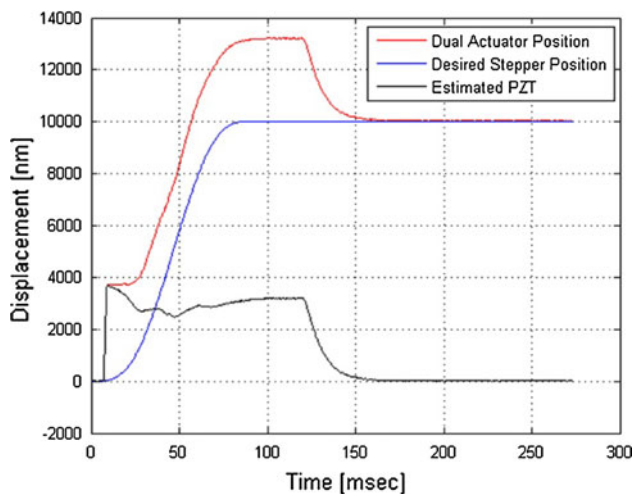


Fig. 15 Experiment results of the conventional algorithm when the reference distance was 10 μm

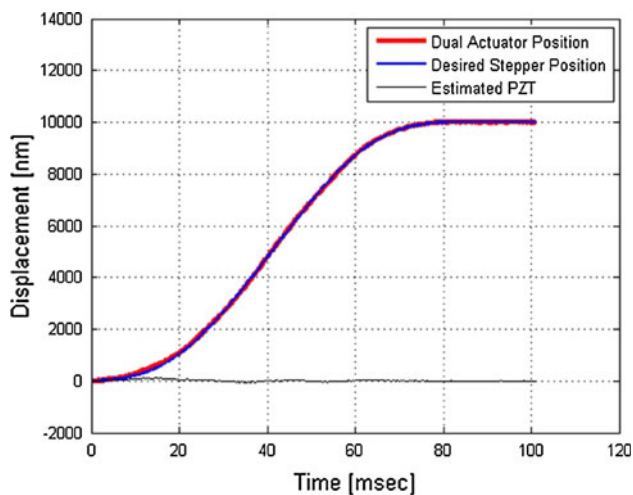


Fig. 16 Experiment results of the proposed algorithm when the reference distance was 10 μm

The reference distance was 500 nm. Figure 14 shows the estimated position of the PZT actuator when the reference distance was 500 nm. The experiment results show good agreement with the simulation results. For the previously studied algorithms, the PZT actuator showed unnecessary operation during the transient response because its frequency response is fast and it tracks the error signal of the reference, as shown in Fig. 12. It can be seen from the proposed algorithm that the unnecessary movement of the PZT actuator was greatly reduced by the tracking position error of the stepper motor at each sampling time, as shown in Fig. 13. Figure 14 shows that the PZT actuator is only used for minor-positioning as its main purpose to achieve the desired resolution of system. The movement of the PZT actuator was reduced by 80.8 % when the reference

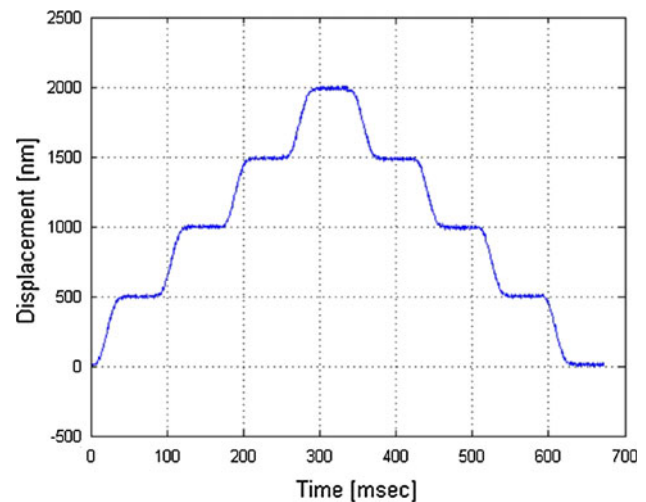


Fig. 17 Experiment results of the staircase response of the single-stage dual-actuator system with 500 nm step

distance was 500 nm. This phenomenon became more obvious when the reference distance exceeded the stroke of the PZT actuator. Figures 15 and 16 show the experiment results of the conventional and proposed algorithms, respectively, when the reference distance was 10 μm . The conventional algorithm results show that the PZT actuator was saturated for 120 ms until the error signal of the reference was below the stroke of the PZT actuator. The proposed algorithm results show that the PZT actuator did not exhibit unnecessary movement during the transient response while successfully tracking the error at a steady state. It can be seen in Figs. 15, 16 and 17 that the proposed algorithm showed a 45.7 % enhanced settling time and a 95.9 % enhanced overshoot when the reference distance was 10 μm . If multiple tasks are given with various desired distance for the manufacturing stage, the proposed algorithm is able to reduce overall manufacturing time by preventing unnecessary PZT motion at the transient response.

6 Conclusions

In this paper, a new mechanical structure of the single-stage dual-actuator system is introduced, and a linear feedback control algorithm preventing unnecessary movement of the PZT actuator at the transient response is proposed for the high-precision manufacturing stage. The estimator and LQR method was used to design the controller. Both the simulation and experiment results show that the proposed algorithm successfully prevents unnecessary motion of the PZT actuator at the transient response while tracking the reference with no steady-state error. The PZT actuator is only used for minor-positioning overall to

achieve the desired resolution of the manufacturing system. The transient response of the system was enhanced by the proposed algorithm. The proposed algorithm showed enhanced overshoot and settling time, especially when the reference distance was greater than the PZT actuator stroke. If multiple tasks are given with various desired distance for the manufacturing stage, the proposed algorithm is able to reduce overall manufacturing time by preventing unnecessary PZT motion at the transient response. The work will be extended to modeling a controller for a non-linear system, and considering the disturbance for the robust controller.

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